

# Lecture #16: State Classification

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# Gambler's Ruin Example

## Example

|   | 0   | 1   | 2   | 3   | 4   |
|---|-----|-----|-----|-----|-----|
| 0 | 1.0 | 0   | 0   | 0   | 0   |
| 1 | 0.6 | 0   | 0.4 | 0   | 0   |
| 2 | 0   | 0.6 | 0   | 0.4 | 0   |
| 3 | 0   | 0   | 0.6 | 0   | 0.4 |
| 4 | 0   | 0   | 0   | 0   | 1.0 |



### Example (Gambler's Ruin)

Suppose for simplicity that  $N = 4$  in Example 1.1, so that the transition probability is

|   | 0   | 1   | 2   | 3   | 4   |
|---|-----|-----|-----|-----|-----|
| 0 | 1.0 | 0   | 0   | 0   | 0   |
| 1 | 0.6 | 0   | 0.4 | 0   | 0   |
| 2 | 0   | 0.6 | 0   | 0.4 | 0   |
| 3 | 0   | 0   | 0.6 | 0   | 0.4 |
| 4 | 0   | 0   | 0   | 0   | 1.0 |

Compute  $p^n$ .

$$p^2 = \begin{pmatrix} 1.00 & 0 & 0 & 0 & 0 \\ 0.60 & 0.24 & 0 & 0.16 & 0 \\ 0.360 & 0.48 & 0 & 0.16 & \\ 0 & 0.36 & 0 & 0.24 & 0.40 \\ 0 & 0 & 0 & 0 & 1.00 \end{pmatrix}, \quad p^{200} \approx \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 57/65 & 0 & 0 & 0 & 8/65 \\ 45/65 & 0 & 0 & 0 & 20/65 \\ 27/65 & 0 & 0 & 0 & 38/65 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$



## Are we guaranteed to return? 0,4

Since  $p(0, 0) = 1$ , it is an absorbing state, the chains come back on the next step with probability 1,

$$\mathbb{P}_0(T_0 = 1) = 1.$$

This means  $\rho_{00} = 1$ , so we know that it is recurrent.

Since  $p(4, 4) = 1$ , it is an absorbing state, the chains come back on the next step with probability 1,

$$\mathbb{P}_4(T_4 = 1) = 1.$$

This means  $\rho_{44} = 1$ , so we know that it is recurrent.



## Can we return? 1,2,3

### Question

*Can we start at 1 and return to 1? (Hint: think about  $p^2(1, 1)$ ).*



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### Solution

$$p^2 = \begin{pmatrix} 1.00 & 0 & 0 & 0 & 0 \\ 0.60 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.40 \\ 0 & 0 & 0 & 0 & 1.00 \end{pmatrix}$$

# Are we guaranteed to return? State 1,3

## Question

*Are we guaranteed to start at 1 and return to 1?*



## Are we guaranteed to return? State 1,3

### Question

*Are we guaranteed to start at 1 and return to 1?*

### Solution

*If the chain goes to 0, it will never return to 1, so the probability of starting at 1 and not returning is*

$$\mathbb{P}_1(T_1 = \infty) \geq p(1, 0) = 0.6 > 0.$$



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### Question

*Are we guaranteed to start at 3 and return to 3?*



## Are we guaranteed to return? State 1,3

### Question

*Are we guaranteed to start at 1 and return to 1?*

### Solution

*If the chain goes to 0, it will never return to 1, so the probability of starting at 1 and not returning is*

$$\mathbb{P}_1(T_1 = \infty) \geq p(1, 0) = 0.6 > 0.$$

### Question

*Are we guaranteed to start at 3 and return to 3?*

### Solution

*If the chain goes to 4, it will never return to 3, so the probability of starting at 3 and not returning is*

$$\mathbb{P}_3(T_3 = \infty) \geq p(3, 4) = 0.4 > 0.$$

# Are we guaranteed to return to 2?

## Question

*Are we guaranteed to start at 2 and return to 2?*



# Are we guaranteed to return to 2?

## Question

*Are we guaranteed to start at 2 and return to 2?*

## Solution

*If the chain goes to 0, it will never return to 2, so the probability of starting at 2, going to 1, going to 0, and not returning is*

$$\mathbb{P}_2(T_2 = \infty) \geq p(2, 1)p(1, 0) = 0.6 \times 0.6 = 0.36 > 0.$$



## Are we guaranteed to return to 2?

### Question

*Are we guaranteed to start at 2 and return to 2?*

### Solution

*If the chain goes to 0, it will never return to 2, so the probability of starting at 2, going to 1, going to 0, and not returning is*

$$\mathbb{P}_2(T_2 = \infty) \geq p(2, 1)p(1, 0) = 0.6 \times 0.6 = 0.36 > 0.$$

### Solution

*If the chain goes to 4, it will never return to 2, so the probability of starting at 2, going to 3, going to 4, and not returning is*

$$\mathbb{P}_2(T_2 = \infty) \geq p(2, 3)p(3, 4) = 0.4 \times 0.4 = 0.16 > 0.$$



## Reaffirming our answer

$$p^2 = \begin{pmatrix} 1.00 & 0 & 0 & 0 & 0 \\ 0.60 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.40 \\ 0 & 0 & 0 & 0 & 1.00 \end{pmatrix}$$



## Social Mobility Example

We recall the transition probability of the social mobility chain:

|               | <b>Lower</b> | <b>Middle</b> | <b>Upper</b> |
|---------------|--------------|---------------|--------------|
| <b>Lower</b>  | 0.7          | 0.2           | 0.1          |
| <b>Middle</b> | 0.3          | 0.5           | 0.2          |
| <b>Upper</b>  | 0.2          | 0.4           | 0.4          |

To determine which states we return to we have to use a slightly different argument.



# Comparing Markov Chains

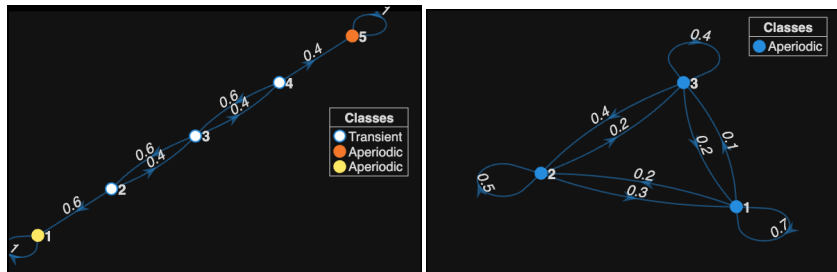


Figure 1: Comparing Chains to understand returning

# Analyzing the Social Mobility Chain

To begin we note that no matter where  $X_n$  is, there is a probability of at least 0.1 of hitting 3 on the next step so



# Definitions

## Definition

Let

be the time of the first return to  $y$  (i.e., being there at time 0 doesn't count) and let

be the probability  $X_n$  returns to  $y$  when it starts at  $y$ .

## Definition

Let

be the time of the  $k^{\text{th}}$  return to  $y$ . Remember starting at  $y$  and returning is  $\rho_{yy}$ , and by the strong Markov property, we know that the time for  $k$  returns is

# Consequences

This means one of two things:

- $\rho_{yy} < 1$ : The probability of returning  $k$  times is  $\rho_{yy}^k \rightarrow 0$  as  $k \rightarrow \infty$ . Thus, eventually the Markov chain does not find its way back to  $y$ . In this case the state  $y$  is called *transient*, since after some point it is never visited by the Markov chain.
- $\rho_{yy} = 1$ : The probability of returning  $k$  times  $\rho_{yy}^k = 1$ , so the chain returns to  $y$  infinitely many times. In this case, the state  $y$  is called *recurrent*, it continually recurs in the Markov chain.



## Definition

We say  $x$  *communicates with*  $y$  and write  $x \rightarrow y$  if there is a positive probability of reaching  $y$  from  $x$ :

## Question

*Can we chain these properties together? What does this say about  $x \rightarrow y$  and  $y \rightarrow z$ ?*

